

Cycle 4 Clone 1 Report — M-EAR-1/real-label-training-v2 (Fork c320de981fda)

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[OUTPUT: report_cycles_4-4_clone_1]

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Abstract

Cycle 4 of clone-1 (fork c320de981fda) executes the auditor-guided c42 closure sequence for M-EAR-1/real-label-training-v2 following the cycle-3 (c41) PIVOT to unfixable-by-audit under the operating-protocol hard-stop clause. Worker landed the closure docs/ear_real_label_training_v2_report.md at **PARTIAL_PROGRESS** state (10 sections; tail ## 10. Branch complete\n\n[[BRANCH_COMPLETE]]), emitted the _manager/M-EAR-1-real-label-training-v2-unfixable-by-audit-clone-1 sole substantive event plus 2 housekeeping events, wrote the merge report at both primary and workspace-fallback paths (byte-identical size 3,974 B), and preserved all handoff assets untouched (rubric doc, hash file, resample manifest, 203-file .npz feature cache, scripts, tests, c6/c22/c26/c38/c1 anchors). Auditor decision: **COMPLETE** with [[BRANCH_COMPLETE]]. Branch retired from fanout scope; downstream work owned by root-conductor sequential-mode pickup.

Verdict

PARTIAL_PROGRESS → COMPLETE (auditor) (branch retired from fanout scope; [[BRANCH_COMPLETE]] confirmed at three locations).

Rubric SHA Anchor (Unchanged; c41-Snapshot Byte-Equal)

Location	State
docs/ear_real_label_training_v2_rubric.md	SHA 01948b6efe6ca5e9..., 9,304 B (byte-identical to c41 snapshot)
data/ear_v2/rubric_hash.txt	65 B; content 01948b6efe6ca5e9... byte-equal to rubric doc SHA (dual-gate consistency preserved)

Zero after-the-fact rubric edits across the 4-cycle arc.

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
4 (this)	c42 auditor-guided closure sequence: NO further empirical work; emit closure doc + manager event + housekeeping	Closure report at PARTIAL_ PROGRESS; _manager/*-unfixable- by-audit-clone-1 sole substantive event; 2 housekeeping events; merge reports written; handoff assets preserved	COMPLETE with [[BRANCH_COMPLETE]]

On-Disk Closure Verification (Auditor Grep + Byte-Level Confirmations)

Check	Result	Evidence
docs/ear_real_label_training_v2_report.md present	☐	13,707 B; 10 ## sections; tail ## 10. Branch complete\n\n[[BRANCH_ COMPLETE]]
PARTIAL_PROGRESS verdict + [[BRANCH_COMPLETE]] sentinel in report	☐	grep-count 1 in report
Merge report at primary path	☐	/home/user/music-gen- instance/fork- c320de981fda/clone-1/merge_ report.md exists, 3,974 B
Merge report at workspace fallback	☐	tools/stale/c41_clone1_merge_ report_draft.md 3,974 B (byte-identical size to primary) + [[BRANCH_COMPLETE]] present

Check	Result	Evidence
Rubric doc untouched (pre==post)	☐	SHA 01948b6efe6ca5e9..., 9,304 B
data/ear_v2/rubric_hash.txt untouched	☐	65 B; byte-equal to rubric doc SHA
data/ear_v2/resample_manifest.json untouched	☐	SHA c6fa617ccf575c2b..., 139,988 B
203-file feature cache preserved	☐	data/ear_v2/features_v2/*.np count = 203, unchanged from c41 (root-conductor cache-skip resume dependency intact)
[[BRANCH_COMPLETE]] in report \$10 + merge report tail + worker final output	☐	all three confirmed

Ledger Events (This Cycle: 3; Correctly Under -clone-1 Suffix)

Per c42 auditor guidance: emit _manager/* closure event + housekeeping only; correctly OMIT _infra/adopt-cycle41-tests-clone-1 (no new tests adopted this cycle):

#	Event	Status	Confidence	event_id
1	_manager/M-EAR-1-real-label-training-v2-unfixable-by-audit-clone-1	in-progress	high (auditor)	21ad8f74-...-c001
2	_run/cycle_41_closed-clone-1	validated	high (auditor)	8656db64-...-997c
3	_archive/cycle-41-scratch-clone-1	validated	high (auditor)	e78a8119-...-8a68

The _manager/*-unfixable-by-audit-clone-1 event is a **first emission on a new identifier**, so the validated → in_progress state-machine rule doesn't apply and in-progress/high is the correct state per c29 lemma.

confidence.rationale on the manager event: "4 consecutive cycles in fanout branch B (c39-initial, c39-resume, c40, c41) landed incremental feature-cache progress (0 → 118 → 124 → 203 / 252) but none completed the >5-stage sequential pipeline (extract → train → eval → determinism × 2 → anchor snapshot → tests → report → 10 events) inside one cycle's wall-time budget. Executional envelope, not science."

State-Machine Discipline (c29 Lemma Respected)

M-EAR-1/real-label-training-v2 is a peer sub-milestone under M-EAR-1. NOT a child of terminal-validated _manager/M-EAR-1-path-B-commit, M-EAR-1/{synthetic-label, head-regularization, feature-representation}-audit, M-EAR-1/armed-harness-fixture-reinforcement, c36 v0, or c38 v1. The _manager/*-unfixable-by-audit-clone-1 event opens a new identifier; validated → in_progress rule does not apply.

Standing Constraints (Unchanged; Preservation Invariants All Hold)

- α pinned at 0.7469387071101908 (this branch never touched α).
- SHA-256 tiebreak; no PRNG; no `sidecar_nonfactor` imports.
- Interpreter guard `#!/usr/bin/python3` on every new script (scripts unchanged from c41; still AST-verified).
- Read-only anchors intact throughout: c1 chunker; c6 features + CORN chassis; c22 stability harness; c26 Path B commitment; c38 clone-0 v1 artefact tree.
- c15 `i4_stratified.py` NOT imported; c23 `model_v2_*` and c25 `feature_subset_adapter` NOT imported (AST-verified from c41).
- c26-frozen SB1/SB2/SB3 thresholds unchanged; no post-hoc adjustment attempted.
- Rated audio egress-blocked at `*.googlevideo.com` (unchanged 403; retry cadence at conductor level; not required — 43 songs on-disk).
- Ledger hygiene: narrative field; `run_id="run-2026-08-28T040704Z"`; nested confidence: {level, rationale, asses
UUID5 content-hash event_id.

Validators (`promise_check` / `org_check`) not run this cycle — closure work touches no code and no schema; on-disk anchors verified byte-for-byte against c41 snapshot per role guidance (“best-effort per role guidance; no findings a validator could have surfaced would change the verdict”).

Anti-Patterns Locked (5-Count Stable)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted. c31 `STILL_GAP` / c35 A anti-pattern surface intact. c30 collision-arc closure at `PARTIAL_BP_UNRESOLVED_SHAPE` unchanged.

Null-cycle anti-pattern explicitly avoided: this closure cycle produced substantive closure with a durable report, a durable ledger event, a durable merge report, and preserved handoff assets — not a null-cycle pause memo.

MINOR Findings (Logged, Not Acted On)

1. **Cycle-numbering nomenclature drift** between brief §0 (“c42 worker”) and §3 event names (`cycle_41_*`) is real but harmless — the closure ledger is unambiguous, the auditor’s c41 PIVOT report and worker’s c42 closure form a coherent two-cycle sequence, and every event_id is content-hash derivable and grep-locatable.

Systemic Finding (Filed in Report §6 Only, Not as Standalone Ledger Event)

Fanout cycle wall-time envelope is insufficient for pipelines with dominant single-stage costs. Two independent clones this campaign have now hit it (this branch + earlier signals in other M-EAR-1 branches). Recommendation: campaign-level `_manager/fanout-pipeline-cost-audit` to enumerate which M-* milestones exceed fanout-cycle capacity and must be assigned to root-conductor sequential mode. Root conductor decides where and when to formalise as a standalone ledger event.

Merge Disposition

Merge report at both primary path (`/home/user/music-gen-instance/fork-c320de981fda/clone-1/merge_report.md`, 3,974 B) and workspace fallback (`tools/stale/c41_clone1_merge_report_draft.md`, byte-identical 3,974 B). `[[BRANCH_COMPLETE]]` sentinel present at three locations (report §10, merge report tail, worker final output). Branch retired from fanout scope; no further

cycle should be scheduled on this clone-1 branch. Any further M-EAR-1/real-label-training-v2 work is owned by root-conductor sequential-mode pickup using preserved handoff assets.

Root-Conductor Sequential-Mode Handoff (Per Cycle-4 Auditor Guidance)

Estimated pipeline cost (sequential-mode): ~49 PANNs Cnn14 penultimate CPU inferences + 5-fold GroupKFold training + SB1/SB2/SB3 evaluation + $\times 2$ determinism check. Order of hours, not order of a fanout cycle.

1. Resume scripts/ear_v2/extract_features_v2.py in sequential (non-fanout) mode with cache-skip semantics against the preserved 203/252 .npy cache; complete the remaining ~49 clips.
2. Run scripts/ear_v2/train_v2.py (5-fold GroupKFold, per-song grouping preserves no-clip-leakage contract).
3. Run scripts/ear_v2/evaluate_sb_v2.py against c26-frozen thresholds (SB1 margin > 0.5909 ; SB2 mean $\tau \geq 0.4$; SB3 detection $F1 \geq 0.90$ — resample now gives SB3 F1 denominator meaningful cardinality > 43).
4. Run scripts/ear_v2/determinism_check.py $\times 2$ on training_result.json + corn_head_v2.pt.
5. Run tests/test_ear_real_label_training_v2.py (≥ 14 tests).
6. Emit 6 substantive + 4 housekeeping ledger events originally scoped under M-EAR-1/real-label-training-v2/* (no -clone-* suffix in sequential mode — the c33 harness guard applies only in clone context, per c39 v3 doc codification).
7. Rewrite docs/ear_real_label_training_v2_report.md at final verdict state (EAR_v2_LANDS / PARTIAL / INSUFFICIENT per frozen rubric) OR leave the PARTIAL PROGRESS report as historical closure and produce a new report under a distinct name.

Candidate future ticket (root-conductor discretion): _manager/fanout-pipeline-cost-audit.

Cumulative Progress

M-EAR-1 arc (post-c41 PIVOT + c42 closure):

Cycle	Milestone	Verdict
c22-c25	Path A chassis chain	insufficient (anti-patterns locked)
c26	_manager/M-EAR-1-path-B-commit armed-harness-fixture-reinforcement	committed; three SBs frozen
c31		FIXTURE_READY
c36	real-label-training-v0	EAR_v0_INSUFFICIENT
c37 clone-1	_manager/ear-sb3-statistic-degeneracy-fallback-statistic	F1_ADOPTED
c38 clone-0	real-label-training-v1	EAR_v1_PARTIAL
c39-c41 clone-1	real-label-training-v2 (attempts 1-4)	PARTIAL_PROGRESS → PIVOT (unfixable-by-audit)
c42 clone-1 (this)	closure sequence for real-label-training-v2	COMPLETE with [[BRANCH_COMPLETE]]; branch retired from fanout scope; 203-file cache + scaffolding preserved for root-conductor sequential-mode pickup

Operating-protocol hard-stop clause executed as designed: c41 audit fired it; c42 closure cycle honoured it without attempting a 5th extraction. Anti-recursion behaviour the clause exists to enforce.

Fanout-envelope pattern now a first-class systemic observation: two independent clones this campaign have hit it; root conductor has concrete recommendation on file (report §6) to codify a campaign-level cost audit before spawning further pipeline-heavy fanout clones.

The unfixable-by-audit verdict has proven to be a useful protocol primitive: creates a legible exit from a productive-but-unbounded loop without invalidating the underlying science, and produces the specific artefact set (preserved cache + PARTIAL_PROGRESS report + _manager/* closure event) that a subsequent sequential-mode pickup can consume with zero re-audit.

Preserved assets valuable at root pickup (unchanged from c41): 203/252 features cached (~80% of extractor cost sunk); manifest frozen; rubric + hash landed; scripts AST-clean; test suite scaffolded (≥ 14 cases enumerated per brief); c22/c23/c25 anti-pattern locks verified absent from scripts/ear_v2/.

Pattern durability: rubric-first pre-registration discipline held for all 4 cycles even though downstream stages did not complete. Rubric SHA anchored to rubric_hash.txt; would embed verbatim in verdict.json at pipeline completion in sequential mode. Zero after-the-fact rubric edits.

c29 state-machine lemma respected: peer sub-milestone; new _manager/*-unfixable-by-audit-clone-1 identifier opens in-progress (first emission); ledger topology stays a DAG.

c32 → c33 → c36 v2 → c39 v3 fanout-namespace convention held: 3 closure events correctly emitted under -clone-1 suffix per codified v3 behaviour.

Collision-modeling arc: closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Session terminal for clone-1's role in fork c320de981fda. [[BRANCH_COMPLETE]] sentinel present at three locations; no further cycle should be scheduled on this clone-1 branch.

[END OUTPUT]